

Théo Matricon

AI Software Engineer — LLMs — Python — Knowledge Systems

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Summary

Computer Science PhD and software engineer specialized in **AI-assisted software systems**, **code generation**, and **machine learning**. Experienced building Python/PyTorch systems, large-scale data pipelines, and rigorous evaluations; interested in **LLMs**, **agents**, **retrieval**, and **internal tooling** that help teams find context and automate workflows.

Technical Skills

AI / ML: LLMs, agentic coding systems, retrieval-augmented generation, structured evaluation, PyTorch

Engineering: Python, SQL, C/C++, Cython, Java, Git, Linux, testing, data pipelines, performance optimization

Algorithms: Program synthesis, code generation, search algorithms, formal methods

Experience

Researcher

INRIA, Rennes, France

Postdoctoral Researcher – AI for Software Engineering

January 2025 – Present

- **Agent Analytics:** Built a large-scale mining pipeline to analyze commits, files, and pull requests across **200M+ GitHub repositories**, studying adoption patterns of **AI coding agents**.
- **Evaluation Infrastructure:** Developed algorithms that reduced large-scale benchmarking costs by **50% on average** while preserving reliable comparisons for fast-moving software systems.
- **Performance Engineering:** Achieved a **4×+ speedup** on a state-of-the-art MBA synthesizer by optimizing its CUDA implementation, balancing algorithmic design and low-level execution constraints.
- **Large-Scale Systems:** Deployed and evaluated software on 50+ node clusters using **200k+ CPU hours**, with reproducible experiment design and automated analysis.
- **Collaboration:** Supervise 3 interns and advise 5 junior researchers in a written, research-driven environment requiring autonomous planning, feedback, and knowledge sharing.

Junior Researcher

University of Bordeaux / LaBRI, Bordeaux, France

PhD Researcher – Program Synthesis & Code Generation

October 2021 – December 2024

- **Research Area:** Designed intelligent software systems combining **search algorithms**, **formal methods**, and **machine learning**.
- **Scaling Algorithms:** Developed a constant-delay search algorithm that ran **2× faster** than prior methods and **scaled exponentially better** with search-space complexity.
- **Software Design:** Designed and maintained DeepSynth2 (20,000+ lines of code), a Python/PyTorch framework for implementing, benchmarking, and comparing AI-based program-synthesis algorithms.
- **Knowledge Integration:** Combined learned program synthesis with external knowledge sources to solve problems beyond conventional synthesizers; related techniques were subsequently applied to **code generation for Excel**.
- **Development:** Implemented performance-critical and ML-driven components in **Python, C/C++, Cython, PyTorch, Java, and SQL**.
- **Leadership:** Published **10+ peer-reviewed papers**, supervised 7+ interns and 2 research engineers, and taught more than 200+ hours.

Education

PhD in Computer Science

Program Synthesis and Code Generation

University of Bordeaux, France

2021 – 2024

Engineering Degree

Engineering Degree / Master's in Computer Science – Artificial Intelligence

ENSEIRB-MATMECA, Bordeaux, France

2018 – 2021